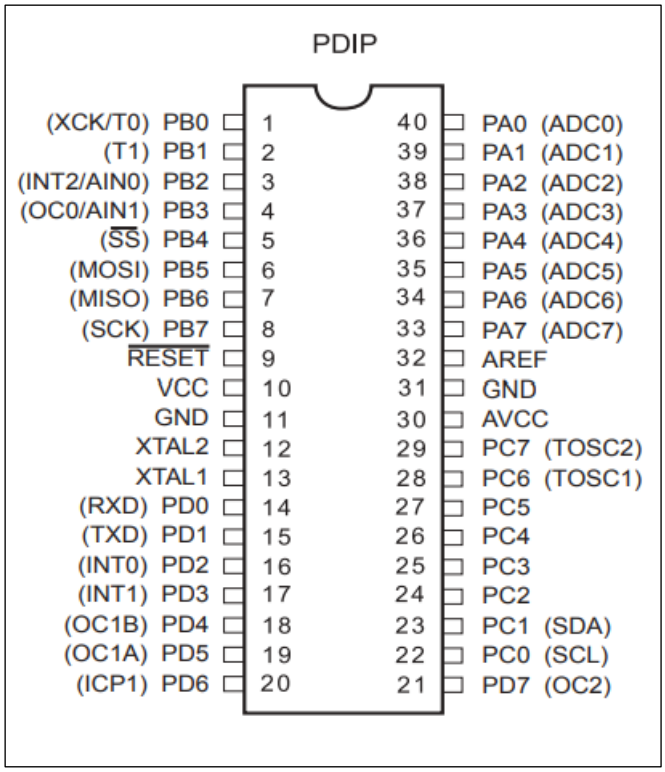
Michael Galileo Hasian (201344020)	**Rangkuman Minggu ke-4**	Senin, 10 September 2022
		Aplikasi Mikrokontroler
3-TNK		Ferry Satria, BSEE, M.T. Rahmawati Hasanah, S.ST.,M.T.

1. Interrupts

Terdapat 3 jenis, INT0, INT1, dan INT2



- GICR (General Interrupt Control Register)

Bit	7	6	5	4	3	2	1	0	
	INT1	INT0	INT2	–	–	–	IVSEL	IVCE	GICR
Read/Write	R/W	R/W	R/W	R	R	R	R/W	R/W	
Initial Value	0	0	0	0	0	0	0	0	

GICR berfungsi untuk memilih External Interrupts mana saja yang akan dipakai dengan cara :

Bit 7 untuk meng-enable INT1

Bit 6 untuk meng-enable INT0

Bit 5 untuk meng-enable INT2

Bit 1, IVSEL, jika IVSEL diberi bit 0 maka Interrupt Vectors ditempatkan di awal dari Flash Memory. Jika IVSEL diberi bit 1 maka Interrupt Vectors dipindahkan ke awal dari Boot Loader section dari Flash. Alamat awal Boot Flash section ditentukan oleh BOOTSZ Fuses.

Bit 0, IVCE (Interrupt Vector Change Enable), IVCE harus diberi bit 1 jika ingin meng-enable perubahan bit IVSEL.

- MCUCR (MCU Control Register)

Bit	7	6	5	4	3	2	1	0	
	SM2	SE	SM1	SM0	ISC11	ISC10	ISC01	ISC00	MCUCR
Read/Write	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	
Initial Value	0	0	0	0	0	0	0	0	

ISC11	ISC10	Description
0	0	The low level of INT1 generates an interrupt request.
0	1	Any logical change on INT1 generates an interrupt request.
1	0	The falling edge of INT1 generates an interrupt request.
1	1	The rising edge of INT1 generates an interrupt request.

ISC01	ISC00	Description
0	0	The low level of INT0 generates an interrupt request.
0	1	Any logical change on INT0 generates an interrupt request.
1	0	The falling edge of INT0 generates an interrupt request.
1	1	The rising edge of INT0 generates an interrupt request.

- MCUCSR (MCU Control and Status Register)

Bit	7	6	5	4	3	2	1	0	
	–	ISC2	–	–	WDRF	BORF	EXTRF	PORF	MCUCSR
Read/Write	R/W	R/W	R	R/W	R/W	R/W	R/W	R/W	
Initial Value	0	0	0	See Bit Description					

Jika ISC2 diisi dengan bit nol, maka transisi negatif akan mengaktifkan *interrupts*, sedangkan jika ISC2 diisi dengan bit satu, maka transisi positif akan mengaktifkan *interrupts.*

- SReg

Status Register (SREG)

SREG	Status Register
C	Carry Flag
Z	Zero Flag
N	Negative Flag
V	Two's Complement Overflow Flag
S	Sign Flag
H	Half Carry Flag
T	Transfer Bit
I	Global Interrupt Enable Bit

Bit	7	6	5	4	3	2	1	0	
	I	**T**	**H**	**S**	**V**	**N**	**Z**	**C**	**SREG**
Read/Write	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	
Initial Value	0	0	0	0	0	0	0	0	

1. I, Global Interrupt Enable, Jika I = 0, Meng-disable semua External Interrupts
2. T, Bit Copy Storage, tempat penyimpanan bit dari register (BST) dan mengambil bit dari T ke register (BLD). Contoh :
   BST R2,3 → Menyimpan bit ke 3 dari R2 menuju T
   BLD R2,7 → Mengambil bit dari T menuju R2 bit ke 7
3. H, Half Carry Flag, bila terdapat carry dari bit ke-3 menuju bit ke-4 maka H = 1
4. S, Sign Bit, merupakan hasil XOR dari S dan N, S = N $\oplus$ V
5. V, Two Complement Overflow Flag, Jika V = 1 maka terjadi overflow
6. N, Negative Flag
7. Z, Zero Flag
8. C, Carry/Borrow Flag, C = 1 jika terdapat carry dari bit ke-7 menuju bit ke-8

Tabel Interrupt

Vector No.	Program Address(2)	Source	Interrupt Definition
1	0x000(1)	RESET	External Pin, Power-on Reset, Brown-out Reset and Watchdog Reset
2	0x001	INT0	External Interrupt Request 0
3	0x002	INT1	External Interrupt Request 1
4	0x003	TIMER2 COMP	Timer/Counter2 Compare Match
5	0x004	TIMER2 OVF	Timer/Counter2 Overflow
6	0x005	TIMER1 CAPT	Timer/Counter1 Capture Event
7	0x006	TIMER1 COMPA	Timer/Counter1 Compare Match A
8	0x007	TIMER1 COMPB	Timer/Counter1 Compare Match B
9	0x008	TIMER1 OVF	Timer/Counter1 Overflow
10	0x009	TIMER0 OVF	Timer/Counter0 Overflow
11	0x00A	SPI, STC	Serial Transfer Complete
12	0x00B	USART, RXC	USART, Rx Complete
13	0x00C	USART, UDRE	USART Data Register Empty
14	0x00D	USART, TXC	USART, Tx Complete
15	0x00E	ADC	ADC Conversion Complete
16	0x00F	EE_RDY	EEPROM Ready
17	0x010	ANA_COMP	Analog Comparator
18	0x011	TWI	Two-wire Serial Interface
19	0x012	INT2	External Interrupt Request 2
20	0x013	TIMER0 COMP	Timer/Counter0 Compare Match
21	0x014	SPM_RDY	Store Program Memory Ready

2. Port pada ATMega 8535

ATMega 8535 memiliki 4 port

- Port A
- Port B
- Port C
- Port D

Masing-masing memiliki nilai 8-bit, sehingga totalnya terdapat 32-bit.

Setiap port dapat difungsikan sebagai port input maupun output (I/O Port).

Port dalam pemrograman ditulisnya PORTA (Port A).

DDRX (Data Direction Register)

Berfungsi untuk menentukan arah dari port (arah keluar = output / arah masuk = input) dengan menggunakan data yang dikirimkan ke register tersebut.



Untuk menggunakannya :

Sebagai Output Port = Data 1

Sebagai Input Port = Data 0

Contoh :

PORT A	Bit-7	Bit-6	Bit-5	Bit-4	Bit-3	Bit-2	Bit-1	Bit-0
	Output				Input			
DDRA	1	1	1	1	0	0	0	0

Untuk DDRX digunakan sebagai output maka dimasukkan data $FF → Port X ← Output

Untuk DDRX digunakan sebagai input maka dimasukkan data $00 → Port X ← Input

3. Contoh Program Penggunaan Port ATMega 8535 Sebagai I/O

1. Buat BCD *Up Counter* dan *Down Counter* 0 – 9. Gunakan 7-*segment Common Anode*.
   a. *Up Counter* 0 - 9

- **Kode Program**

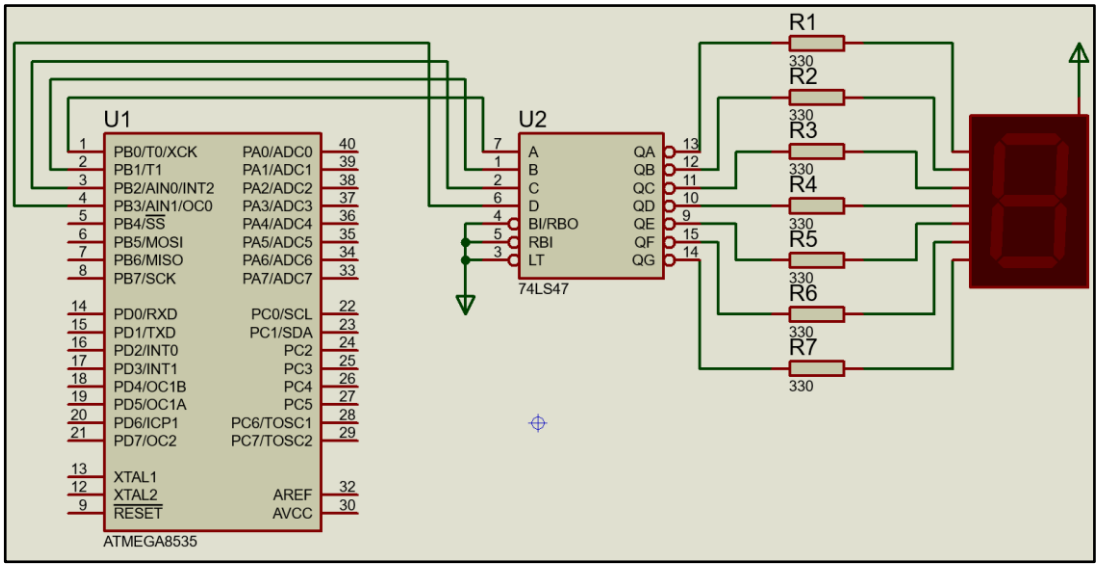
```
.INCLUDE"M8535DEF.INC"
.ORG 0
RJMP MAIN
        ; DECIMAL UPCOUNTER, DENGAN 74LS47
        ; DELAY MENGGUNAKAN LOOP 3 REGISTER
        ;IMPLEMENTASI: PROTEUS DEC_COUNT_01 [DI
PROTEUS_PROJECT]

MAIN:       LDI R16,LOW(RAMEND)
            OUT SPL,R16
            LDI R16,HIGH(RAMEND)
            OUT SPH,R16
```

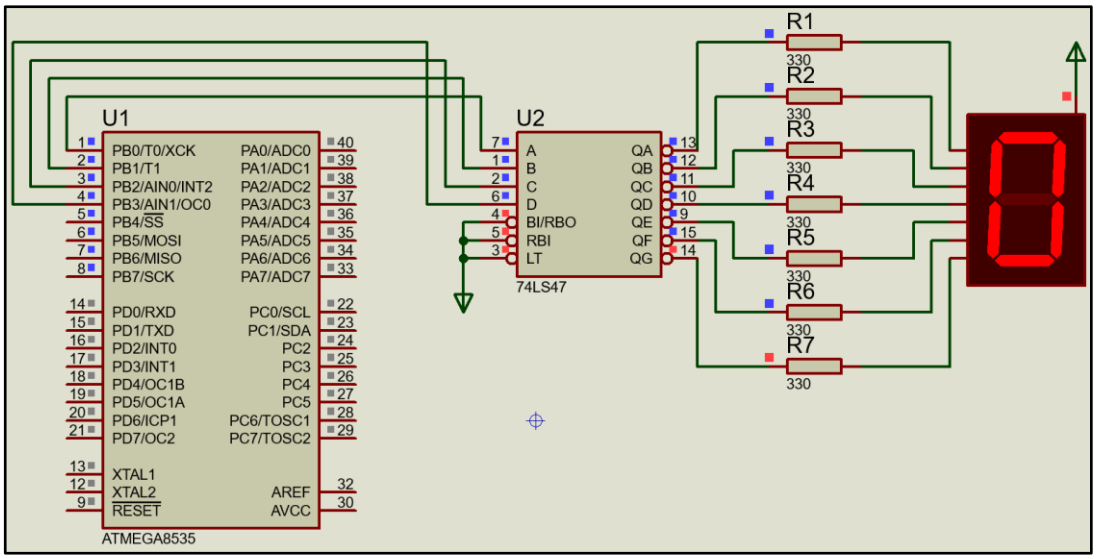
```
        LDI R20,0XFF
        OUT DDRB,R20 ; PORT B SBG OUTPUT
ITU:    LDI R18,0
SANA:   OUT PORTB,R18
        RCALL DELAY
        INC R18          ; Menaikkan nilai counter
        CPI R18,0X0A     ; Membandingkan nilai counter dengan 10
        BRNE SANA        ; Jump jika < 10
        RJMP ITU         ; Mengulang counter dari awal

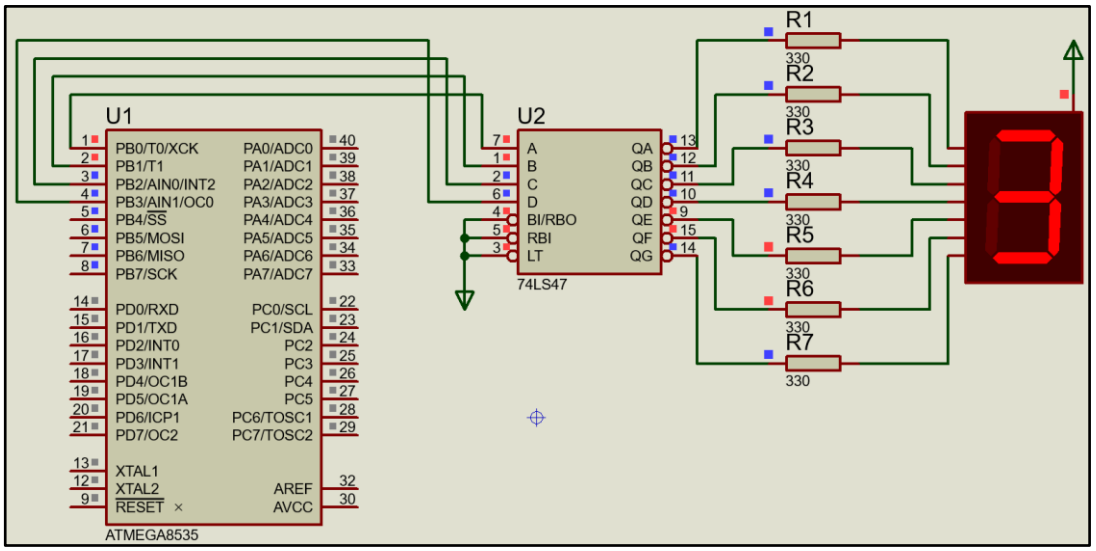
DELAY:  LDI R23,3
LUP1:   LDI R24,0XFF
LUP2:   LDI R25,0
LUP3:   DEC R25
        BRNE LUP3
        DEC R24
        BRNE LUP2
        DEC R23
        BRNE LUP1
        RET
```

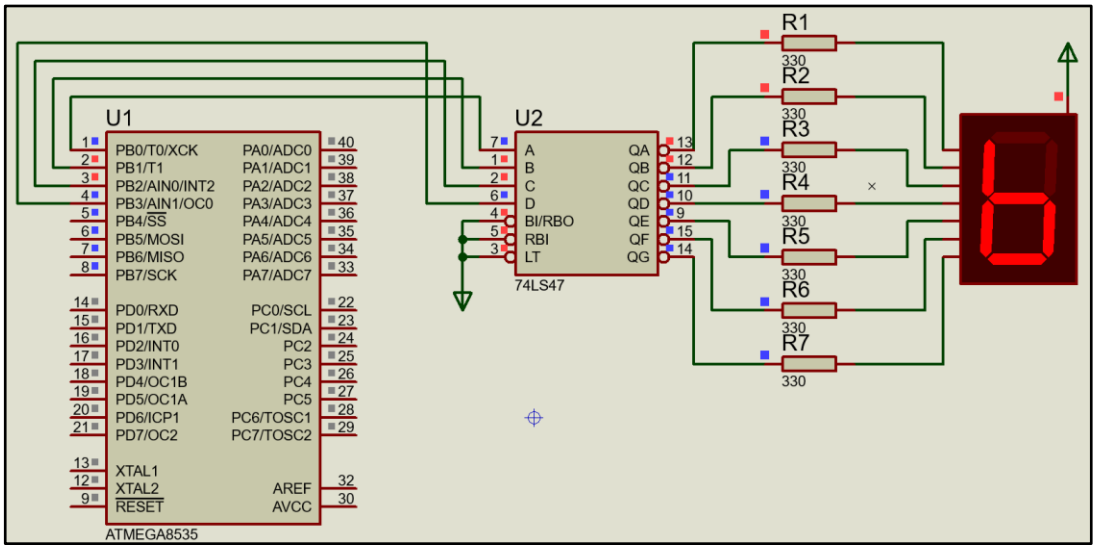
- **Skema Rangkaian**

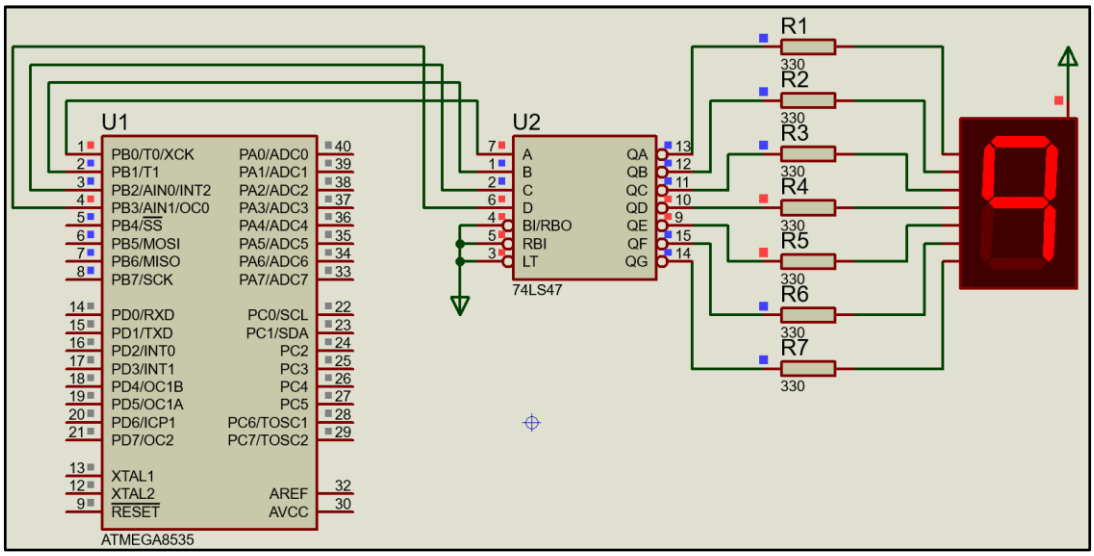


- **Hasil Simulasi Rangkaian**


U1
PB0/T0/XCK
PB1/T1
PB2/AIN0/INT2
PB3/AIN1/OC0
PB4/SS
PB5/MOSI
PB6/MISO
PB7/SCK
PD0/RXD
PD1/TXD
PD2/INT0
PD3/INT1
PD4/OC1B
PD5/OC1A
PD6/ICP1
PD7/OC2
XTAL1
XTAL2
RESET
PA0/ADC0
PA1/ADC1
PA2/ADC2
PA3/ADC3
PA4/ADC4
PA5/ADC5
PA6/ADC6
PA7/ADC7
PC0/SCL
PC1/SDA
PC2
PC3
PC4
PC5
PC6/TOSC1
PC7/TOSC2
AREF
AVCC
ATMEGA8535
U2
A
B
C
D
BI/RBO
RBI
LT
QA
QB
QC
QD
QE
QF
QG
74LS47
R1
R2
R3
R4
R5
R6
R7
330


U1
PB0/T0/XCK
PB1/T1
PB2/AIN0/INT2
PB3/AIN1/OC0
PB4/SS
PB5/MOSI
PB6/MISO
PB7/SCK
PD0/RXD
PD1/TXD
PD2/INT0
PD3/INT1
PD4/OC1B
PD5/OC1A
PD6/ICP1
PD7/OC2
XTAL1
XTAL2
RESET
PA0/ADC0
PA1/ADC1
PA2/ADC2
PA3/ADC3
PA4/ADC4
PA5/ADC5
PA6/ADC6
PA7/ADC7
PC0/SCL
PC1/SDA
PC2
PC3
PC4
PC5
PC6/TOSC1
PC7/TOSC2
AREF
AVCC
ATMEGA8535
U2
A
B
C
D
BI/RBO
RBI
LT
QA
QB
QC
QD
QE
QF
QG
74LS47
R1
R2
R3
R4
R5
R6
R7
330


U1
PB0/T0/XCK
PB1/T1
PB2/AIN0/INT2
PB3/AIN1/OC0
PB4/SS
PB5/MOSI
PB6/MISO
PB7/SCK
PD0/RXD
PD1/TXD
PD2/INT0
PD3/INT1
PD4/OC1B
PD5/OC1A
PD6/ICP1
PD7/OC2
XTAL1
XTAL2
RESET
PA0/ADC0
PA1/ADC1
PA2/ADC2
PA3/ADC3
PA4/ADC4
PA5/ADC5
PA6/ADC6
PA7/ADC7
PC0/SCL
PC1/SDA
PC2
PC3
PC4
PC5
PC6/TOSC1
PC7/TOSC2
AREF
AVCC
ATMEGA8535
U2
A
B
C
D
BI/RBO
RBI
LT
QA
QB
QC
QD
QE
QF
QG
74LS47
R1
R2
R3
R4
R5
R6
R7
330



b. *Down Counter* 0 - 9

- **Kode Program**

```
.INCLUDE"M8535DEF.INC"
.ORG 0
RJMP MAIN
        ; DECIMAL DOWNCOUNTER, DENGAN 74LS47
        ; DELAY MENGGUNAKAN LOOP 3 REGISTER
        ; IMPLEMENTASI: PROTEUS DEC_COUNT_01 [DI
PROTEUS_PROJECT]

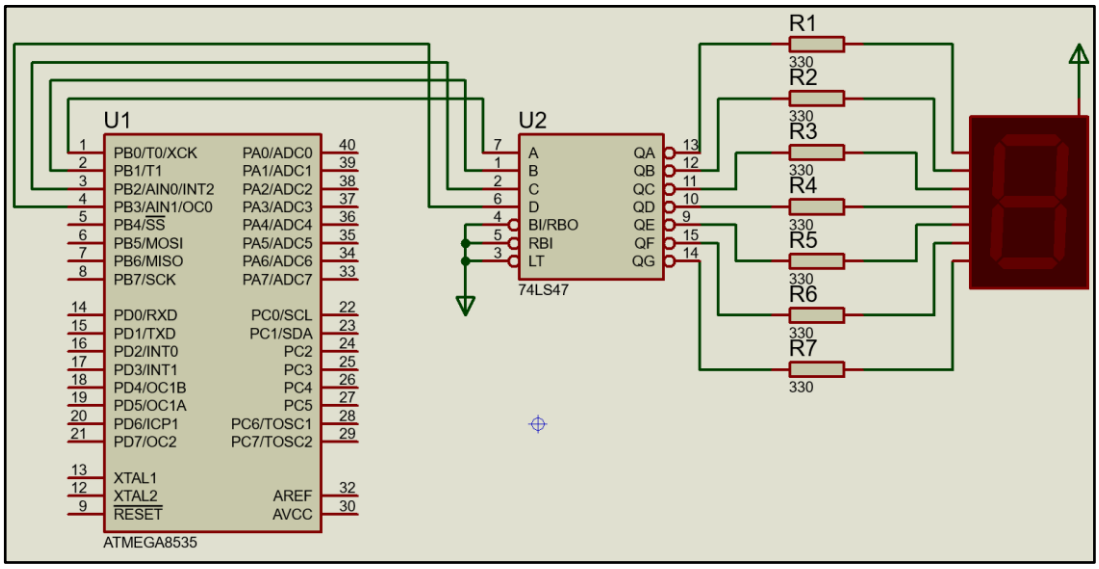
MAIN:       LDI R16,LOW(RAMEND)
            OUT SPL,R16
            LDI R16,HIGH(RAMEND)
            OUT SPH,R16

            LDI R20,0XFF
            OUT DDRB,R20 ; PORT B SBG OUTPUT
ITU: LDI R18,9              ; input nilai awal counter
SANA:       OUT PORTB,R18
            RCALL DELAY
            DEC R18             ; mengurangi counter
            CPI R18,1           ; membandingkan dengan 1
```

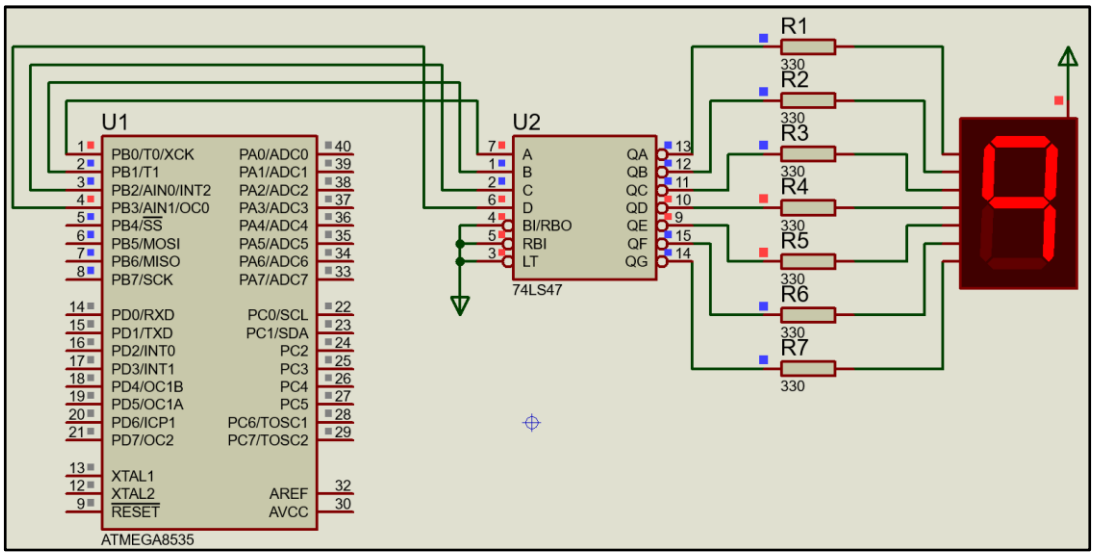
```
            BRNE SANA       ; counter lanjut jika < 1
            RJMP ITU        ; mengulangi counter

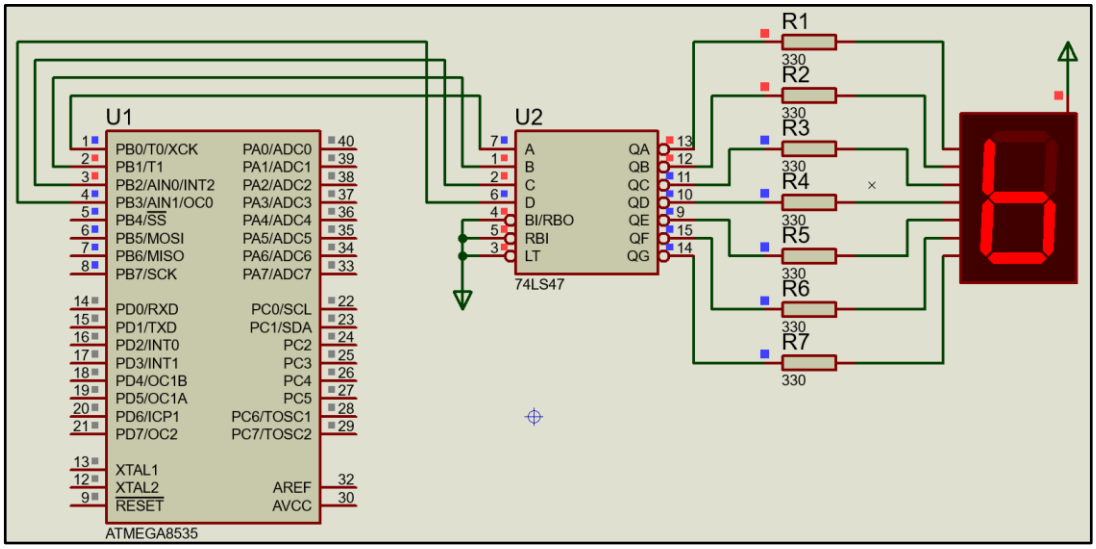
DELAY:      LDI R23,3
LUP1:       LDI R24,0XFF
LUP2:       LDI R25,0
LUP3:       DEC R25
            BRNE LUP3
            DEC R24
            BRNE LUP2
            DEC R23
            BRNE LUP1
            RET
```

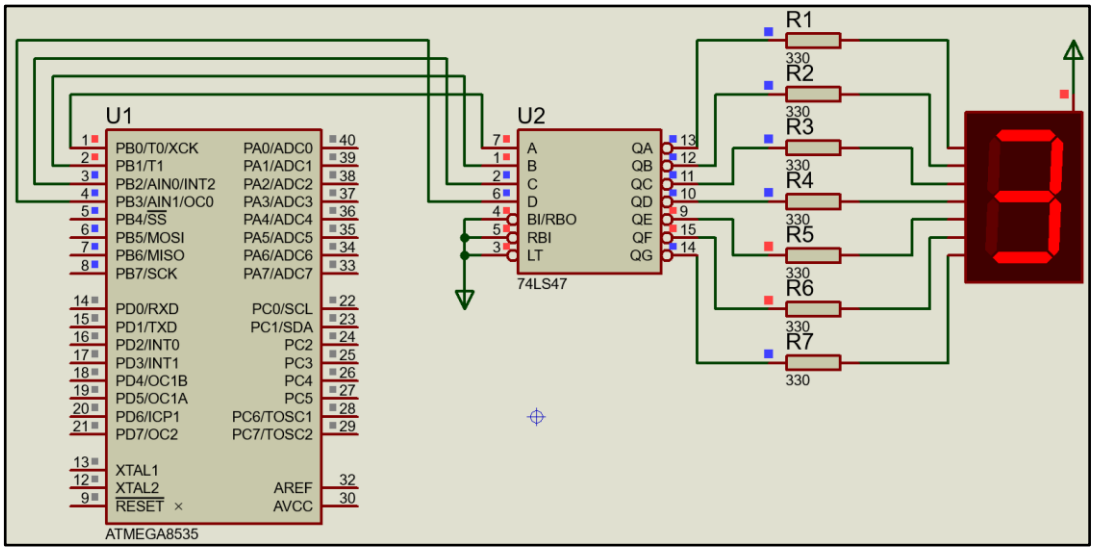
- **Skema Rangkaian**

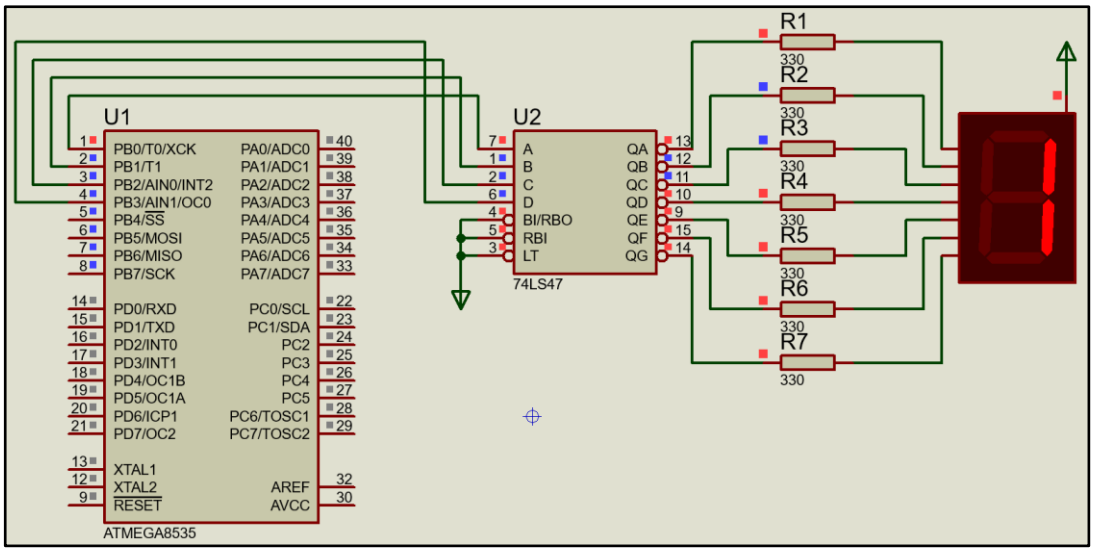


- **Hasil Simulasi Rangkaian**


U1
PB0/T0/XCK
PB1/T1
PB2/AIN0/INT2
PB3/AIN1/OC0
PB4/SS
PB5/MOSI
PB6/MISO
PB7/SCK
PD0/RXD
PD1/TXD
PD2/INT0
PD3/INT1
PD4/OC1B
PD5/OC1A
PD6/ICP1
PD7/OC2
XTAL1
XTAL2
RESET
PA0/ADC0
PA1/ADC1
PA2/ADC2
PA3/ADC3
PA4/ADC4
PA5/ADC5
PA6/ADC6
PA7/ADC7
PC0/SCL
PC1/SDA
PC2
PC3
PC4
PC5
PC6/TOSC1
PC7/TOSC2
AREF
AVCC
ATMEGA8535
U2
A
B
C
D
BI/RBO
RBI
LT
QA
QB
QC
QD
QE
QF
QG
74LS47
R1
R2
R3
R4
R5
R6
R7
330


U1
PB0/T0/XCK
PB1/T1
PB2/AIN0/INT2
PB3/AIN1/OC0
PB4/SS
PB5/MOSI
PB6/MISO
PB7/SCK
PD0/RXD
PD1/TXD
PD2/INT0
PD3/INT1
PD4/OC1B
PD5/OC1A
PD6/ICP1
PD7/OC2
XTAL1
XTAL2
RESET
PA0/ADC0
PA1/ADC1
PA2/ADC2
PA3/ADC3
PA4/ADC4
PA5/ADC5
PA6/ADC6
PA7/ADC7
PC0/SCL
PC1/SDA
PC2
PC3
PC4
PC5
PC6/TOSC1
PC7/TOSC2
AREF
AVCC
ATMEGA8535
U2
A
B
C
D
BI/RBO
RBI
LT
QA
QB
QC
QD
QE
QF
QG
74LS47
R1
R2
R3
R4
R5
R6
R7
330


U1
PB0/T0/XCK
PB1/T1
PB2/AIN0/INT2
PB3/AIN1/OC0
PB4/SS
PB5/MOSI
PB6/MISO
PB7/SCK
PD0/RXD
PD1/TXD
PD2/INT0
PD3/INT1
PD4/OC1B
PD5/OC1A
PD6/ICP1
PD7/OC2
XTAL1
XTAL2
RESET
PA0/ADC0
PA1/ADC1
PA2/ADC2
PA3/ADC3
PA4/ADC4
PA5/ADC5
PA6/ADC6
PA7/ADC7
PC0/SCL
PC1/SDA
PC2
PC3
PC4
PC5
PC6/TOSC1
PC7/TOSC2
AREF
AVCC
ATMEGA8535
U2
A
B
C
D
BI/RBO
RBI
LT
QA
QB
QC
QD
QE
QF
QG
74LS47
R1
R2
R3
R4
R5
R6
R7
330



2. Buat BCD *Up Counter Modulo* 26 (0-25). Gunakan 7-*segment Common Anode*.

- **Kode Program**

```
.INCLUDE"M8535DEF.INC"
.ORG 0
RJMP MAIN
                ; DECIMAL UPCOUNTER MODULO 26, DENGAN 74LS47
                ; DELAY MENGGUNAKAN LOOP 3 REGISTER
          ; IMPLEMENTASI: PROTEUS DEC_COUNT_01 [DI
PROTEUS_PROJECT]

MAIN:       LDI R16,LOW(RAMEND)
            OUT SPL,R16
            LDI R16,HIGH(RAMEND)
            OUT SPH,R16

            LDI R20,0XFF
            OUT DDRB,R20      ; PORT B SBG OUTPUT
ITU: LDI R18,0              ; input awal counter
SANA:       OUT PORTB,R18
            RCALL DELAY
            LDI R20,1
            ADD R18,R20                 ; menaikkan nilai counter
```

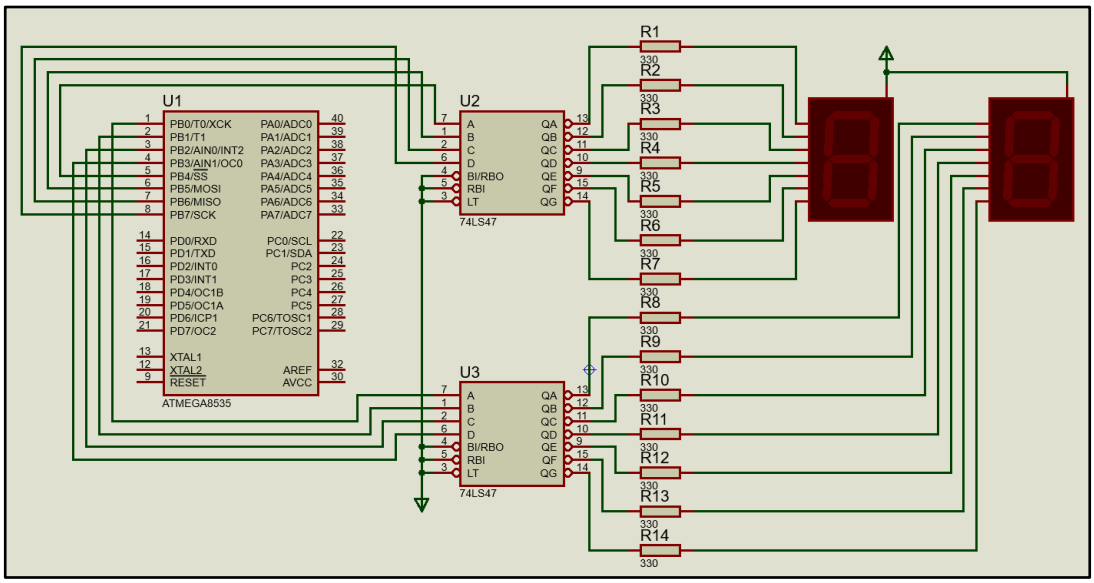
```
                MOV R19,R18
                RCALL DAA       ; menambahkan faktor koreksi DAA
                MOV R18,R19
                CPI R18,0X26    ; membandingkan dengan 26
                BRLT SANA       ; counter lanjut jika < 26
                RJMP ITU        ; mengulang counter

DELAY:  LDI R23,3
LUP1:   LDI R24,0XFF
LUP2:   LDI R25,0
LUP3:   DEC R25
                BRNE LUP3
                DEC R24
                BRNE LUP2
                DEC R23
                BRNE LUP1
                RET
daa:  push r16;amankan isi reg. ke stack
                push r17
                push r21
                push r24
                in r16,sreg; amankan sreg
                in r17,sreg
                ldi r24,0 ; r24 hanya sbg status dr carry flag
                adc r24,r24; status carry flag sebelum koreksi
; ambil, amankan kembali sreg, uji half carry
                out sreg,r17
                brhc lup4; lompat ke lup4 jika H=0
                rcall fk_ln; satuan + 06 jika H=1
        ;       rjmp lup5
lup4: mov r21,r19
                andi r21,$0f; ambil dgt.satuan
                cpi r21,$0a ; > 9
                brcs lup5; lompat ke lup5 jika < 9
```

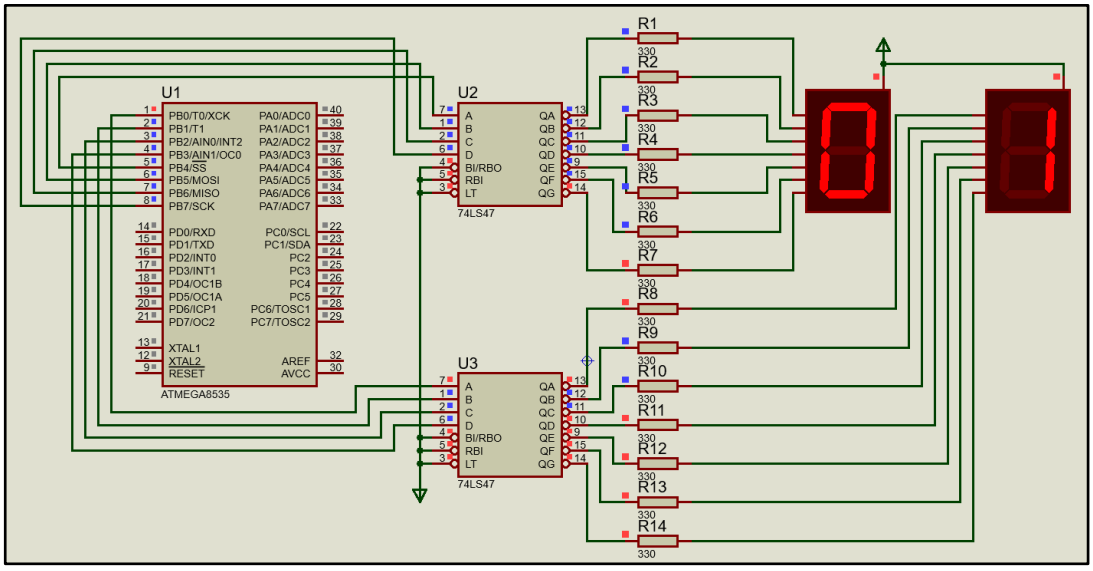
```
                rcall fk_ln; satuan + 06 jika > 9
lup5: out sreg,r17; ambil, amankan kembali sreg, uji carry
                brcc lup6; lompat ke lup6 jika C=0
                rcall fk_hn; puluhan + 60 jika C=1
        ;       rjmp lup7
lup6: mov r21,r19
                andi r21,0xf0; ambil dgt.puluhan
                cpi r21,0xa0
                brcs lup7; lompat ke lup7 jika pul < a
                rcall fk_hn; pul + 60 jika > 9
lup7: out sreg,r16; mengatur carry flag sblm keluar:
                clc; C = 0
                in r16,sreg ;amankan sreg stlh c=0
                cpi r24,0 ;menguji status
                breq lup8
                out sreg,r16;ambil sreg, set c
                sec; C = 1
                in r16,sreg; amankan sreg stlh c=1
lup8:
                out sreg,r16;status akhir
                pop r24;kembalikan isi reg. dr stack
                pop r21
                pop r17
                pop r16
                ret
fk_ln:          ldi r21,06; koreksi dgt.satuan
                out sreg,r16
                add r19,r21
                in r16,sreg
                brcc tuh
                ldi r21,0x01
                add r17,r21
tuh:   out sreg,r16
                adc r24,r24
```

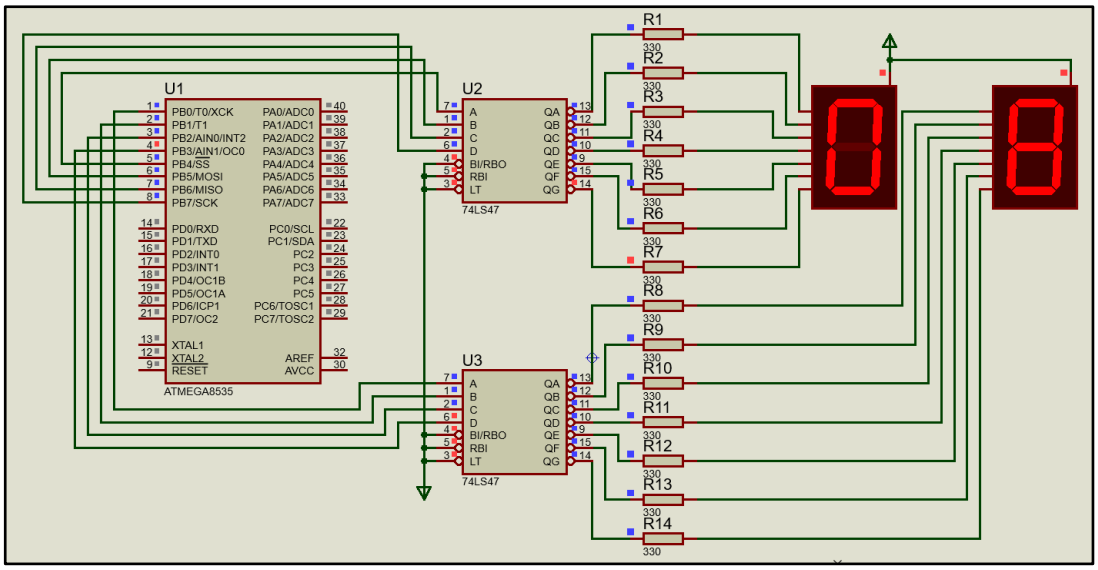
```
                ret
fk_hn:          ldi r21,$60; koreksi dgt.puluhan
                out sreg,r16
                add r19,r21
                in r16,sreg
                adc r24,r24; amankan carry pd r24
                ret ; akhir dari prog. subrutin.
```

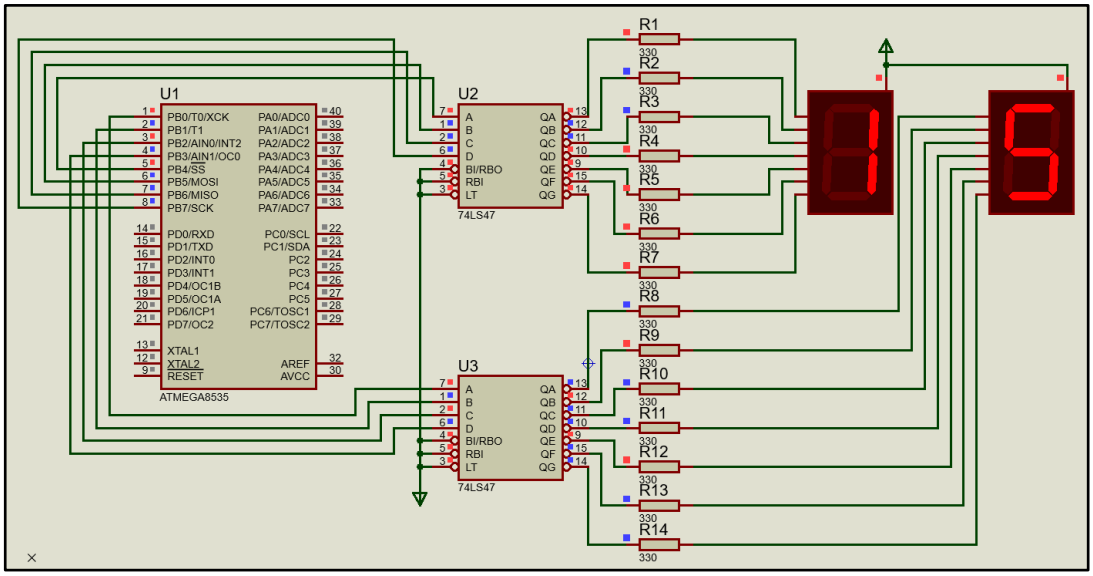
- **Skema Rangkaian**

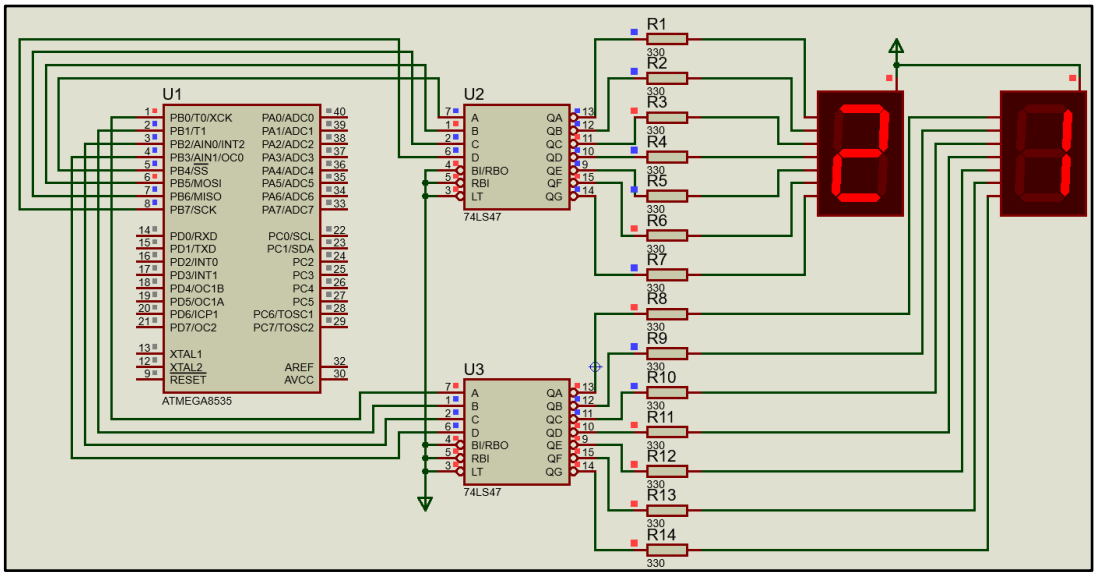


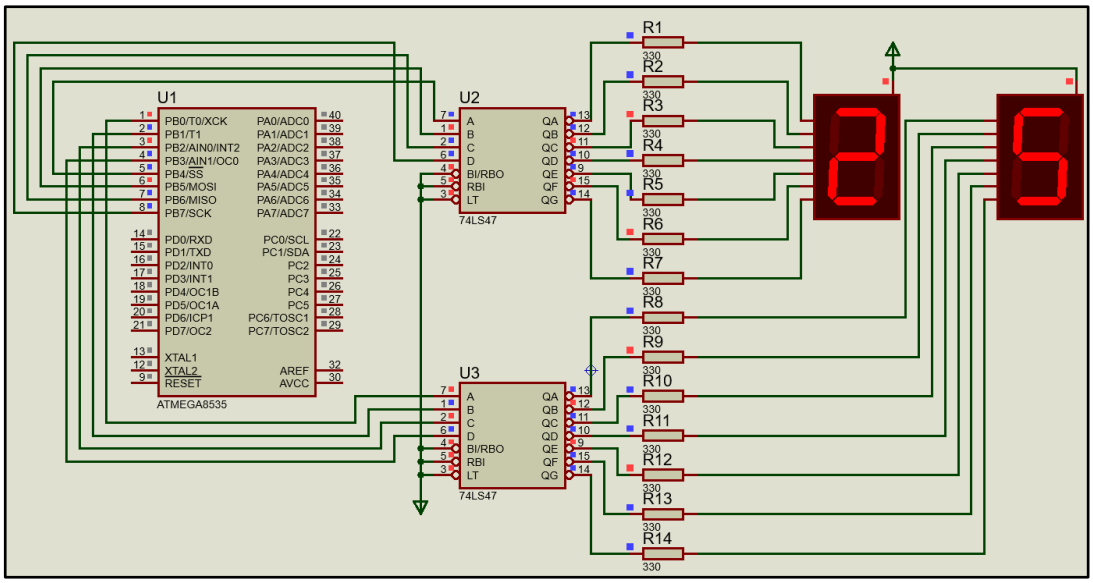
- **Hasil Simulasi Rangkaian**




U1
ATMEGA8535
U2
U3
74LS47
R1
R2
R3
R4
R5
R6
R7
R8
R9
R10
R11
R12
R13
R14
330


U1
ATMEGA8535
U2
U3
74LS47
R1
R2
R3
R4
R5
R6
R7
R8
R9
R10
R11
R12
R13
R14
330


U1
ATMEGA8535
U2
U3
74LS47
R1
R2
R3
R4
R5
R6
R7
R8
R9
R10
R11
R12
R13
R14
330



3. Buat BCD *Down Counter Modulo* 31 (31-0). Gunakan 7-*segment Common Anode*.

- **Kode Program**

```
.INCLUDE"M8535DEF.INC"
.ORG 0
RJMP MAIN
            ; BCD DOWN COUNTER MODULO 31, DENGAN 74LS47
            ; DELAY MENGGUNAKAN LOOP 3 REGISTER
        ; IMPLEMENTASI: PROTEUS DEC_COUNT_01 [DI
PROTEUS_PROJECT]

MAIN:       LDI R16,LOW(RAMEND)
            OUT SPL,R16
            LDI R16,HIGH(RAMEND)
            OUT SPH,R16

            LDI R20,0XFF
            OUT DDRB,R20      ; PORT B SBG OUTPUT
ITU: LDI R18,0X31         ; input nilai awal counter
SANA:       OUT PORTB,R18
            RCALL DELAY
            LDI R20,0X99          ; komplemen 10 dari 1 = 99
            ADD R18,R20                ; sama dengan pengurangan dengan 1
```

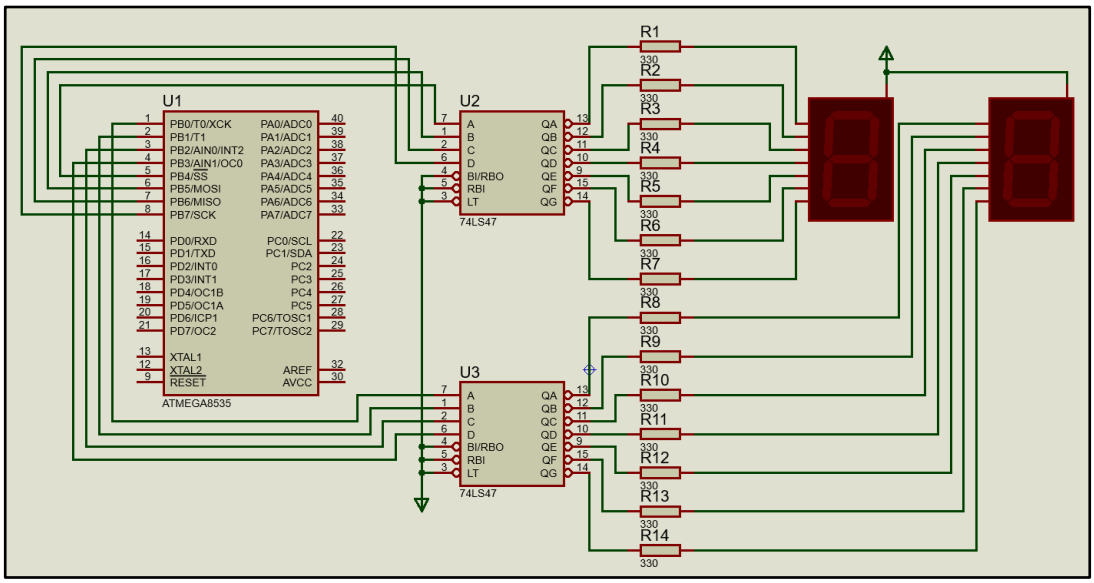
```
                MOV R19,R18
                RCALL DAA       ; penambahan faktor koreksi DAA
                MOV R18,R19
                CPI R18,0X99    ; dibandingkan dengan k'10 dari 1
                BRNE SANA       ; jump jika > 1
                RJMP ITU

DELAY:  LDI R23,3
LUP1:   LDI R24,0XFF
LUP2:   LDI R25,0
LUP3:   DEC R25
                BRNE LUP3
                DEC R24
                BRNE LUP2
                DEC R23
                BRNE LUP1
                RET
daa:  push r16;amankan isi reg. ke stack
                push r17
                push r21
                push r24
                in r16,sreg; amankan sreg
                in r17,sreg
                ldi r24,0 ; r24 hanya sbg status dr carry flag
                adc r24,r24; status carry flag sebelum koreksi
; ambil, amankan kembali sreg, uji half carry
                out sreg,r17
                brhc lup4; lompat ke lup4 jika H=0
                rcall fk_ln; satuan + 06 jika H=1
        ;       rjmp lup5
lup4: mov r21,r19
                andi r21,$0f; ambil dgt.satuan
                cpi r21,$0a ; > 9
                brcs lup5; lompat ke lup5 jika < 9
```

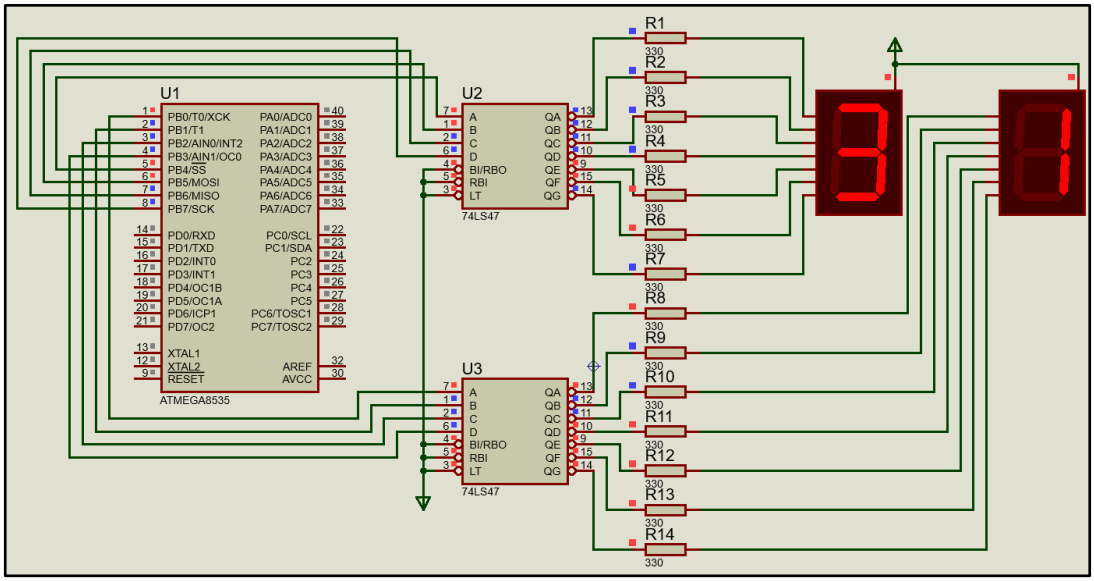
```
                rcall fk_ln; satuan + 06 jika > 9
lup5: out sreg,r17; ambil, amankan kembali sreg, uji carry
                brcc lup6; lompat ke lup6 jika C=0
                rcall fk_hn; puluhan + 60 jika C=1
        ;       rjmp lup7
lup6: mov r21,r19
                andi r21,0xf0; ambil dgt.puluhan
                cpi r21,0xa0
                brcs lup7; lompat ke lup7 jika pul < a
                rcall fk_hn; pul + 60 jika > 9
lup7: out sreg,r16; mengatur carry flag sblm keluar:
                clc; C = 0
                in r16,sreg ;amankan sreg stlh c=0
                cpi r24,0 ;menguji status
                breq lup8
                out sreg,r16;ambil sreg, set c
                sec; C = 1
                in r16,sreg; amankan sreg stlh c=1
lup8:
                out sreg,r16;status akhir
                pop r24;kembalikan isi reg. dr stack
                pop r21
                pop r17
                pop r16
                ret
fk_ln:          ldi r21,06; koreksi dgt.satuan
                out sreg,r16
                add r19,r21
                in r16,sreg
                brcc tuh
                ldi r21,0x01
                add r17,r21
tuh:   out sreg,r16
                adc r24,r24
```

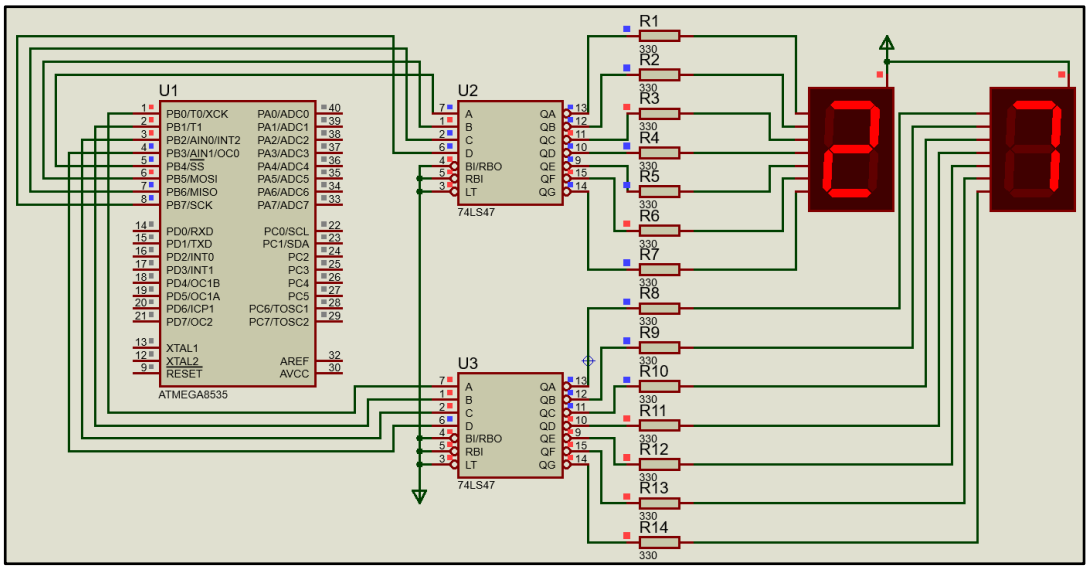
```
                ret
fk_hn:          ldi r21,$60; koreksi dgt.puluhan
                out sreg,r16
                add r19,r21
                in r16,sreg
                adc r24,r24; amankan carry pd r24
                ret ; akhir dari prog. subrutin.
```

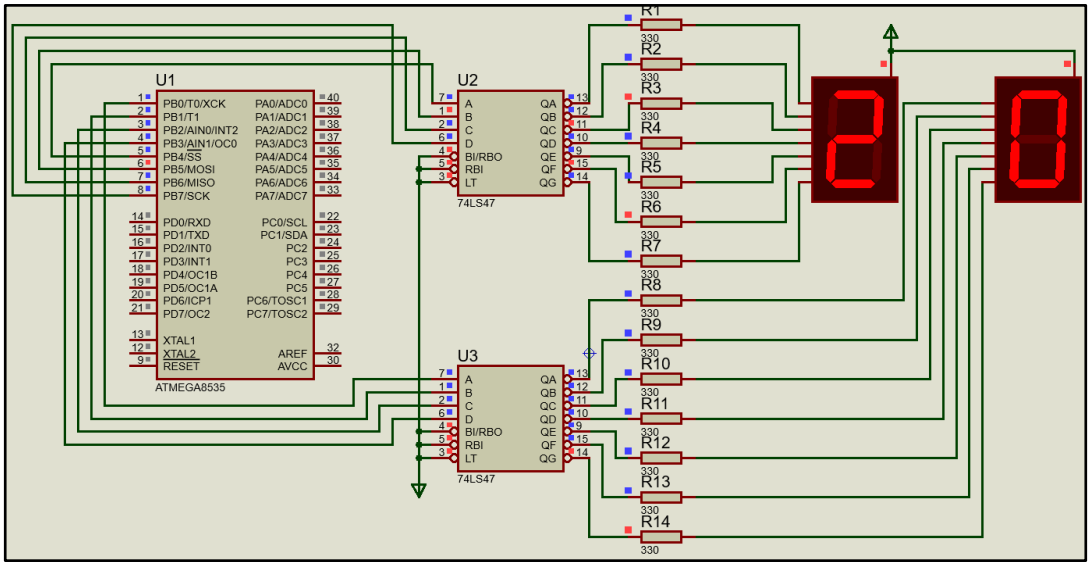
- **Skema Rangkaian**

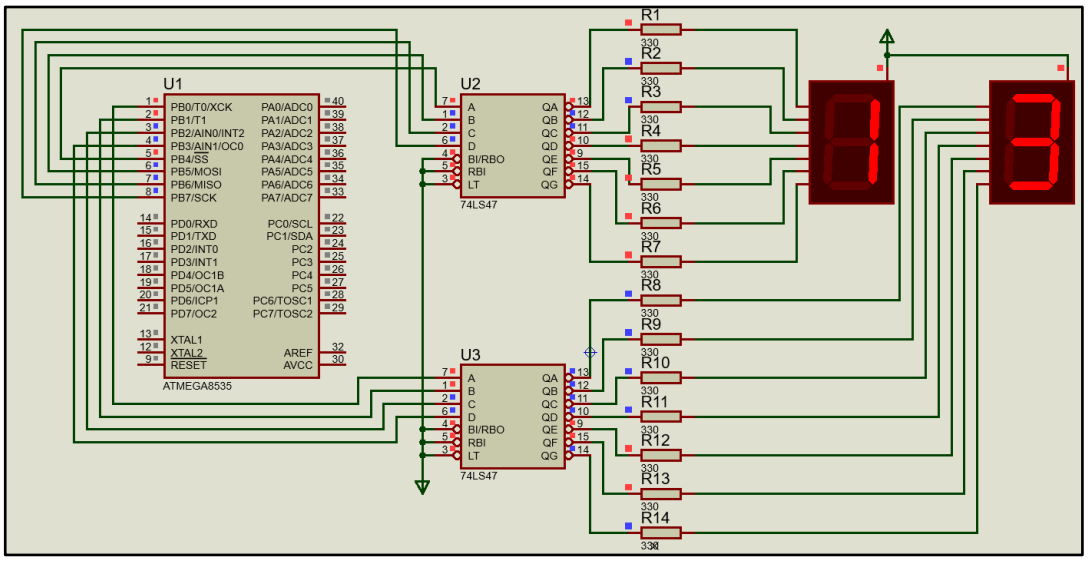


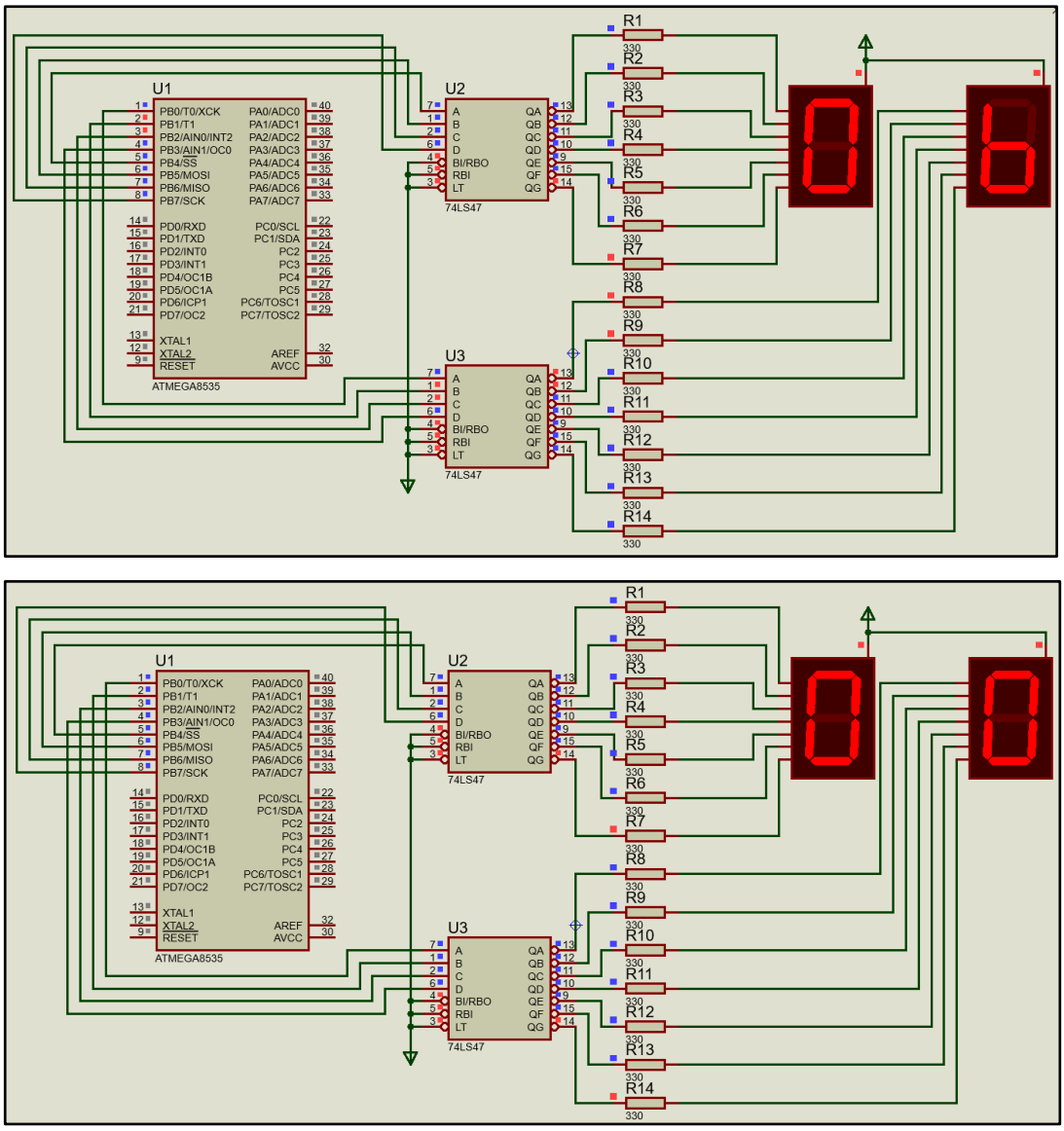
- **Hasil Simulasi Rangkaian**




U1
ATMEGA8535
U2
U3
74LS47
R1
R2
R3
R4
R5
R6
R7
R8
R9
R10
R11
R12
R13
R14
330


U1
ATMEGA8535
U2
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R1
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330


U1
ATMEGA8535
U2
U3
74LS47
R1
R2
R3
R4
R5
R6
R7
R8
R9
R10
R11
R12
R13
R14
330



4. Buat BCD *Up*/*Down Counter Modulo* 36 (0-35). Mode *up*/*down* dikendalikan oleh *push button*. Gunakan 7-*segment Common Cathode*.
   - **Kode Program**
   - **Skema Rangkaian**
   - **Hasil Simulasi Rangkaian**